# Electromagnet_V1 First experiment Phase 3 Thermal test Calculations

resistance of 28 AWG wire $\approx 0.213\ \Omega/m$

28 AWG insulated diameter: $\approx 0.35$ mm

Core diameter: 8mm

600 turns

length of core: 20mm

turns per layer =

$\frac{20}{0.35} \approx 57$ turns/layer

$\#$ of layers $= 600/57 \approx 10.5 \approx 11$

first layer $= 8+0.35 = 8.35$ mm

eleventh layer $= 8+21(0.35) = 15.35$mm

average diameter $\approx \frac{8.35+15.35}{2} = 11.85$

average length per turn $\approx \pi(11.85) \approx 37.2$

total wire length $= 600 \cdot 37.2$

$\approx 22.3$m

Resistance $\approx (0.213\ \Omega/m \cdot 22.3m)$

$$\boxed{R_0 \approx 4.75\ \Omega}$$

For voltage:

$V = IR_0$

Once we evaluate the lower amperages we can find a more accurate $R_{th}$

We must derive A thermal equation.

## Heat Stored:

$Q = C_{th}\Delta T$ where $C_{th} = \sum m_i c_i$

$C_{th} = m_{cu}c_{cu} + m_{steel}c_{steel} \longrightarrow c_{cu} \approx 385\ J/(kg°C)$, $c_{steel} \approx 490\ J/(kg°C)$

$\frac{dQ}{dt} = C_{th}\frac{dT}{dt}$

## Electrical heat entering:

$P_{in} = I^2R$

and we're assuming it's constant

$P_{in} = p$

## Heat leaving:

$P_{loss} = H(T - T_{amb})$

$R_{th} = \frac{1}{H}$

$P_{loss} = \frac{T - T_{amb}}{R_{th}}$

Create differential equation:

$$C_{th}\frac{dT}{dt} = p - \frac{T - T_{amb}}{R_{th}}$$

$\theta(t) = T(t) - T_{amb}$

Since $T_{amb}$ is assumed const

$\frac{d\theta}{dt} = \frac{dT}{dt}$

$$C_{th}\frac{d\theta}{dt} = p - \frac{\theta(t)}{R_{th}}$$

$\frac{d\theta}{dt} + \frac{\theta}{C_{th}R_{th}} = \frac{p}{C_{th}}$

Solve homogeneous

$\frac{d\theta}{dt} + \frac{\theta}{C_{th}R_{th}} = 0$

$k = \frac{1}{C_{th}R_{th}}$

$\frac{d\theta}{dt} + k\theta = 0$

$\frac{d\theta}{dt} = -k\theta$

$\frac{1}{\theta}\frac{d\theta}{dt} = -k$

$\frac{d\theta}{\theta} = -k\,dt$

$\ln|\theta| = -kt + C_1$

$|\theta| = e^{C_1}e^{-kt}$

$\rightarrow |\theta| = Ce^{-kt}$

$\theta_h(t) = Ce^{\frac{-t}{R_{th}C_{th}}}$

$\frac{\theta_p}{R_{th}C_{th}} = \frac{p}{C_{th}}$

So general solution is

$\theta(t) = RR_{th} + Ce^{-t/(R_{th}C_{th})}$

$T_0 = T_{ambient}$

$\theta(0) = 0$

$0 = RR_{th} + C = 0$

$C = -pR_{th}$

$\theta = T - T_{amb}$

giving us:

$$\boxed{T(t) = T_{amb} + pR_{th}\left(1 - e^{-t/(R_{th}C_{th})}\right)}$$

m of Steel:

$V = \pi r^2 L$

$r = 0.004$m $\quad L = 0.035$m

$= 1.759 \cdot 10^{-6}\ m^3$

$\rho_{steel} \approx 7850\ kg/m^3$

$$\boxed{m_{steel} \approx 0.0138kg}$$

$m_{cu}$:

$d_{cu} \approx 0.321$mm

$A_{cu} = \pi\left(\frac{d}{2}\right)^2 \approx 8.09 \cdot 10^{-8}\ m^2$

$V_{cu} = A_{cu}L_{wire}$

$V_{cu} = 1.80 \cdot 10^{-6}\ m^3$

$\rho_{cu} \approx 8960\ kg/m^3$

$m_{cu} = (8960)(1.80 \cdot 10^{-6}) \approx 0.0162kg$

$m_{cu} \approx 16.2g$

$C_{th} = m_{cu}c_{cu} + m_{steel}c_{steel}$

$C_{th} \approx 6.24 + 6.76$

$$\boxed{C_{th} \approx 13.0\ J/°C}$$

unfortunately $R_{th}$ is difficault to calculate thus we'll run at 3 different values,

$$\boxed{R_{th} = 20, 40, 60\ °C/W}$$

Because voltage changes with temperature,

$$V(t) = IR_0\left(1 + 0.00393\left(T(t) - T_0\right)\right)$$